University of Ghana

Department of Statistics and Actuarial Science

ACTU 301: Life Contingencies I

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EXERCISE 2

2.1 A mortality table has a select period of three years.

(i) Find expressions in terms of the life table functions $l_{[x]+t}$ and l_y for

(a) $q_{[50]}$

(b) ${}_2p_{[50]}$

(c) ${}_{2|}q_{[50]}$

(d) ${}_{2|3}\,q_{[50]+1}$

(ii) Calculate ${}_3p_{53}$ given that:

$$q_{[50]} = 0.01601, \qquad {}_2q_{[50]} = 0.96411,$$
$$ {}_{2|}q_{[50]} = 0.02410, \qquad {}_{2|3}q_{[50]+1} = 0.09272$$

2.2 A certain life table has a select period of 1 year. At each integer age x, the select rate mortality is 50% of the ultimate rate. Calculate $e_{[60]}$, given that e_{60}=17.5424 and q_{60} = 0.0142.

2.3 Given the following excerpt of a life table

x	l_x
50	100,000
51	99,900
52	99,700
53	99,500
54	99,100
55	98,500

Calculate the following:

(a) ${}_2d_{52}$

(b) ${}_{3|}q_{50}$

2.4 Given:

$$l_x = 10000e^{-0.05x}, \qquad x \geq 0.$$

Find ${}_{5|15}q_{10}$.

2.5 Given the following excerpt of a life table

x	l_x
40	10000
41	9900
42	9700
43	9400
44	9000
45	8500

(i) Assuming uniform distribution of deaths between integral ages, calculate the following:

(a) ${}_{0.2}p_{42}$

(b) ${}_{2.6}q_{41}$

(c) ${}_{1.6}q_{40.9}$

(ii) Repeat Question 2.5 (ii) by assuming constant force of mortality between integral ages.

2.6 Given:

(i) $l_{40} = 9{,}313{,}166$
(ii) $l_{41} = 9{,}287{,}264$
(iii) $l_{42} = 9{,}259{,}571$

Assuming uniform distribution of deaths between integral ages, find ${}_{1.4}q_{40.3}$

2.7 Given

x	l_x
40	60500
50	55800
60	50200
70	44000
80	36700

Assuming that deaths are uniformly distributed over each 10-year interval, find ${}_{15|20}q_{40}$

2.8 Given the following select-and-ultimate table with a select period of 2 years:

x	$q_{[x]}$	$q_{[x]+1}$	q_{x+2}	$x+2$
50	0.02	0.04	0.06	52
51	0.03	0.05	0.07	53
52	0.04	0.06	0.08	54

Find ${}_{2|2}q_{[50]}$

2.9 Given the following select-and-ultimate table with a select period of 2 years:

x	$l_{[x]}$	$l_{[x]+1}$	l_{x+2}	$x+2$
70	22507	22200	21722	72
71	21500	21188	20696	73
72	20443	20126	19624	74
73	19339	19019	18508	75
74	18192	17871	17355	76

(a) Compute ${}_3p_{73}$.

(b) Compute the probability that a life age 71 dies between ages 75 and 76, given that the life was selected at age 70.

(c) Assuming uniform distribution of deaths between integral ages, calculate ${}_{0.5}p_{[70]+0.7}$.

(d) Assuming constant force of mortality between integral ages, calculate ${}_{0.5}p_{[70]+0.7}$

2.10 Given

$$f_0(t) = \frac{20-t}{200}, \qquad 0 \le t < 20$$

Find $\mathring{e}_5$

2.11 Given the following select-and-ultimate life table

x	$q_{[x]}$	$q_{[x]+1}$	q_{x+2}	$x+2$
65	0.01	0.04	0.07	67
66	0.03	0.06	0.09	68
67	0.05	0.08	0.12	69

(i) State the select period.

(ii) Calculate ${}_{1|2}q_{[65]+1}$

(iii) Calculate ${}_{0.4}p_{[66]+0.3}$, assuming constant force of mortality between integer ages.

2.12 Given the following life table

x	l_x	x	l_x	x	l_x
91	27	94	12	97	3
92	21	95	8	98	1
93	16	96	5	99	0

(a) Calculate e_{91}.

(b) Calculate $\mathring{e}_{91}$, assuming uniform distribution of deaths between integer ages

2.13 Given the following 4-year select-and-ultimate life table

$[x]$	$q_{[x]}$	$q_{[x]+1}$	$q_{[x]+2}$	$q_{[x]+3}$	q_{x+4}	$x+4$
40	0.00101	0.00175	0.00205	0.00233	0.00257	44
41	0.00113	0.00188	0.00220	0.00252	0.00293	45
42	0.00127	0.00204	0.00240	0.00280	0.00337	46
43	0.00142	0.00220	0.00262	0.00316	0.00384	47
44	0.00157	0.00240	0.00301	0.00367	0.00445	48

(a) Construct the table of $l_{[x]+t}$, for x = 40, 41, 42 and for all *t*.
Use $l_{[40]}$ =10,000.

(b) Calculate the following probabilities:

(i) ${}_{2}p_{[42]}$

(ii) ${}_{3}q_{[41]+1}$

(iii) ${}_{3|2}q_{[41]}$

2.14 Given the following excerpt of a life table

x	l_x
50	100,000
51	99,900
52	99,700
53	99,500
54	99,100
55	98,500

(a) Calculate d_{52}

(b) Calculate $_{2|}q_{50}$

(c) Assuming uniform distribution of deaths between integer ages, calculate the value of $_{4.3}p_{50.4}$

(d) Assuming constant force of mortality between integer ages, calculate the value of $_{4.3}p_{50.4}$

2.15 In a certain non-select mortality table, there is a uniform distribution of deaths between any two consecutive integer ages. Find formulae in terms of l_{30}, l_{31} and l_{32} for

$$f_T(t) = \mu e^{-\mu t},\ t \geq 0.$$

Assume a constant force of interest δ, find $\bar{A}_x$, ${}^2\bar{A}_x$, and $\mathrm{Var}(\bar{Z}_x)$

2.16 In a certain non-select mortality table, there is a uniform distribution of deaths between any two consecutive integer ages. Find formulae in terms of l_{30}, l_{31} and l_{32} for

(i) $_{1.5}p_{30.5}$

(ii) $\mu_{30.5}$

2.17 Suppose that the life table function is

$$l_x = 10{,}000(100-x)^2 \qquad 0 \le x \le 100$$

Identify the cumulative distribution function and the probability density function for the associated lifetime variable *X*.

2.18 Suppose that the force of mortality for a survival model is given by the formula:

$$\mu(x) = \frac{0.9}{90-x} \quad \text{for} \quad 0 \le x < 90$$

Compute the following probabilities:

(a) $_{2.5}p_{20}$

(b) $_{2.5}q_{20}$

(c) $_{2.5|}q_{20}$